



Gitesh Kumar

Machine Learning Engineer |
LLM Systems & Agentic AI |
Production AI

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ML engineer building and industrializing production-grade LLM systems, RAG pipelines and agentic AI workflows. Built and deployed a live industrial agentic AI system end-to-end, from data architecture through zero-trust production deployment and monitoring. AWS Certified Generative AI Developer – Professional. I combine AI engineering with 4+ years of operational experience in industrial environments where reliability and trust are non-negotiable.

CERTIFICATIONS

AWS Certified Generative AI Developer – Professional (AIP-C01)

2026

PROJECTS (OPEN SOURCE)

Industrial Agentic AI System (live)

- Built a full **industrial data pipeline** from scratch. Raw sensor and machine data flows through a Medallion architecture (Bronze → Silver → Gold), cleaned, quality-flagged and enriched with live energy pricing and financial risk scores, with **automated pipeline health monitoring** throughout.
- Developed a **natural language AI agent** (LangChain, Llama 3 70B) that lets engineers and C-level managers ask operational questions in plain English. The agent writes its own SQL, queries a live data lakehouse and **returns grounded answers** with operational recommendations.
- Built three deployment modes. **Fully on-premise** via Ollama for environments where data cannot leave the facility and two cloud modes via Groq for scalability, reflecting the real data sovereignty trade-off in industrial environments.
- Extended the deployment with an **OpenZiti zero-trust overlay network**, providing identity-based access to the API and dashboard in line with industrial **data sovereignty and security** requirements.
- Deployed as a **FastAPI service with a live dashboard** featuring a real-time machine command centre, natural language query interface and pipeline health monitoring, containerised with Docker.
- Live deployment – industrialdatapipeline.up.railway.app/dashboard

DTC Inventory Intelligence (2026)

- Second end-to-end pipeline: Bronze/Silver/Gold architecture on retail transaction data producing **sell-through rates, markdown risk with cost estimates, reorder signals** and inventory health scores across 7 divisions.
- An **LLM agent** on top answers inventory questions in plain English against the Gold layer.

WORK EXPERIENCE

Graduate Researcher

04/2023 - 07/2025 (2 years+)

Institute of Factory Automation and Production Systems (FAPS) Nürnberg, BY, Germany

Mini Thesis: Explainable AI in Computer Vision for End-of-Line Inspection

- Developed an **end-to-end defect detection pipeline** using YOLOv8 for industrial assemblies, specifically addressing the black-box problem in safety-critical quality inspection workflows.
- Implemented Grad-CAM, LRP and LIME to visualize model decision logic, conducting audits to ensure the **AI was identifying actual defects** rather than relying on image artifacts or spurious correlations.
- Balanced detection accuracy against interpretability so the **system could be safely trusted** and adopted by human operators on the shopfloor.

Master's Thesis: A Retrieval-Augmented Framework for Fine-tuned LLMs for Vulnerability Analysis

- Developed and evaluated a retrieval-augmented LLM pipeline using LLaMA, implementing LoRA/QLoRA fine-tuning and FAISS-based vector retrieval for domain-specific knowledge access.
- Conducted systematic evaluation and statistical analysis comparing RAG-enabled and baseline models using NLP metrics to assess reliability in regulated environments.

- Managed day-to-day operations of a high-capacity LPG bottling plant ensuring **reliability of critical systems** and served as primary technical contact with Honeywell on automation projects.
- Drove digital transformation of plant workflows by **integrating SCADA and ERP systems**, eliminating manual reporting and giving all teams real-time visibility into plant performance for the first time.
- Partnered with maintenance and safety teams and other stakeholders to build datasets from operational sensor data that supported production monitoring and helped **identify recurring inefficiencies** before they escalated.
- Developed a KPI cockpit that translated raw machine data into real-time operational insights, giving management full transparency across production performance for the first time and contributing to a **10% improvement in production efficiency**.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg

Masters in Elektromobilität-ACES

Grade: 2.1 | Specialization: AI and Sustainable Production Technology

10/2022 - 03/2026

PEC University of Technology Chandigarh, India

Bachelors of Engineering

Mechanical Engineering

07/2013 - 05/2017

TECHNICAL SKILLS

AI & ML - Python, PyTorch, Scikit-learn, LangChain, RAG (FAISS), LoRA/QLoRA fine-tuning, YOLOv8, Explainable AI (Grad-CAM, LRP, LIME), Agentic AI Architecture, LLM Orchestration, LLM Evaluation & Guardrails, Model Monitoring

GenAI on AWS - Amazon Bedrock, Amazon SageMaker (certified, AIP-C01)

Data Engineering - ETL/ELT Pipelines (Medallion), SQL, SQLite, OPC-UA, SCADA Systems, ERP Integration, Industrial IoT, Real-time Data Streams, Anomaly Detection, Data Quality & Pipeline Health Monitoring

Development - FastAPI, REST APIs, Docker, Git, Railway, OpenZiti (Zero-Trust Networking)

Domain - Industrial AI, Production AI Systems, OT/IT Integration, End-to-End AI Deployment, Cross-functional Delivery

LANGUAGES	SOCIAL NETWORKS	INTERESTS
English IELTS: 8/9 German B1 (actively improving)	GitHub: @gitesh-kumar LinkedIn: Gitesh Kumar	State-level tennis player, homebrewer, philosophy reader, occasional stand-up comedian